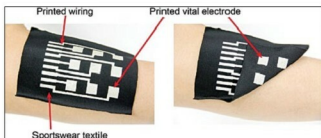


Now, print electronic circuits

Researchers from University of Tokyo have created a mechanism that puts electronics on apparels with a new type of conductive ink. The ink will allow printing of electronic circuits for a wide range of biomedical



Electronics on apparels with a new conductive ink
(Image courtesy: www.sciencedaily.com)

sensors directly on the textile. This means that a shirt would also function as a heart rate monitor.

So far the ink could only be printed directly on clothes to function as electronic circuits. Prof. Takao Someya and his team at Graduate School of Engineering, Tokyo, have succeeded in producing an ink that is elastic and can conduct electricity. It can be printed on textile in just one step. The team used flakes of silver, organic solvents, fluorine rubber and fluorine surfactant in the ink to achieve desirable properties. The end result is the ink that can stretch to 2.5 times its original length.

Spectrometer that fits inside smartphones

Scientists at MIT have created a spectrometer that could enable the devices to diagnose diseases, especially skin conditions, or detect environmental pollutants.

Spectrometers measure the properties of light and are widely used in physical, chemical and biological research. These new spectrometers can fit inside a smartphone camera, using tiny semiconductor nanoparticles called quantum dots.



The quantum dot spectrometer can be printed onto a smartphone camera
(Image courtesy: www.wired.co.uk)

Quantum dots, a type of nanocrystals discovered in early 1980s, are made by combining metals such as lead or cadmium with other elements including sulphur, selenium or arsenic. By controlling the ratio of these materials, temperature and reaction time, scientists can generate a nearly unlimited number of dots with differences in an electronic property known as bandgap, which determines the wavelengths of light that each dot will absorb.

The spectrometer deploys hundreds of quantum dot materials that each filter a specific set of wavelengths of light. Quantum-dot filters are printed into a thin film and placed on top of a photodetector such as the charge-coupled devices (CCDs) found in smartphone cameras.

Robot that can be controlled by thoughts and brain signals

Researchers at Swiss Federal Institute of Technology (EPFL), Switzerland, have developed a robot that can be controlled with thoughts and brain signals. It can perform various tasks, too. The team, headed by Prof. Jose del R. Millain, particularly had disabled people in mind while working on the concept to give them a sense of independence.

Nine disabled and ten healthy people in Italy, Germany and Switzerland took part in the task of piloting a robot with their thoughts. For several weeks, each subject put on an electrode-studded hat capable of analysing his brain signals. They then instructed the robot to move, transmitting their instructions in real-time via the Internet from their home country. By virtue of a video camera, screen and wheels, the robot, located in an EPFL laboratory in Switzerland, was able to film as it moved while displaying the face of the remote pilot via Skype.

In the second part of the tests, disabled people with residual mobility were asked to pilot the robot with movements they were still capable of doing.

Samsung screen shows the road ahead

Large vehicles are often too hard to pass on a narrow two-lane road. Samsung has come up with a clever solution to this problem. Argentinean arm of the South Korean tech giant has launched what it calls a Safety Truck, a semi truck with a wireless camera mounted on the front, displaying the road ahead on a screen tacked on to the back of the truck. The move is a part of an effort to reduce head-on collisions while passing.

The front-mounted camera broadcasts its signal to four monitors on the back of the truck to give drivers behind the truck a good view ahead. In addition to making passing safer, Samsung says that this would let drivers see any obstacles on the road ahead, preventing the need for sudden emergency braking.